Enterprise AI Platform

Disease Prediction & Clinical Diagnostics Report

Task Type: Classification

Automated ML Experiment Report

Generated by CodeAlpha AutoML Pipeline

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1. Executive Summary

This report presents a comprehensive evaluation of **5** machine learning models for a **classification** task.

The top-performing model is **XGBoost** with an accuracy / score of **0.8525**.

The dataset contains **303** samples and **14** features.

All models were trained, evaluated, and ranked using the CodeAlpha automated ML pipeline with SHAP-based explainability.

2. Dataset Overview

Property	Value
Total Rows	303
Total Cols	14
Target Variable	target (Heart Disease Present)
Class Distribution	Normal: 138 (45.5%), Diseased: 165 (54.5%)
Features Analyzed	age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, th

3. Preprocessing Pipeline

The following preprocessing steps were applied to the raw dataset before model training:

Step 1: Target encoding and logical column validation
Step 2: Missing values imputation using median strategies
Step 3: Scaling numerical metrics (cholesterol, blood pressure) via StandardScaler
Step 4: One-Hot encoding categorical variables (cp, thal, slope)

4. Model Architectures

Model	Architecture / Description
SVM	Support Vector Machine classifier with RBF kernel and optimized C/gamma coefficients.
XGBoost	Extreme Gradient Boosting decision trees trained with binary logistic loss.
Random Forest	Bagged ensemble of 100 decision trees with Gini impurity split criteria.
CatBoost	Categorical Gradient Boosting with symmetric trees and built-in categorical feature handling.
MLP	Multi-Layer Perceptron neural network with 2 hidden layers (64, 32 nodes) and ReLU activat

5. Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	Roc Auc	Inference Ms	Training Sec
XGBoost	0.8525	0.8485	0.8615	0.8549	0.9120	1.2000	4.5000
CatBoost	0.8492	0.8421	0.8580	0.8500	0.9054	1.8000	8.2000
Random Forest	0.8410	0.8350	0.8532	0.8440	0.8985	2.1000	3.1000
SVM	0.8250	0.8180	0.8360	0.8268	0.8804	0.8000	1.5000
MLP	0.8120	0.8050	0.8240	0.8144	0.8712	1.5000	12.4000

6. Leaderboard

Models are ranked by a weighted composite score: Accuracy×0.4 + Precision×0.2 + Recall×0.2 + (1/inference_ms)×0.1 + (1/training_sec)×0.1

Rank	Model	Weighted Score
1	**XGBoost**	**0.8525**
2	CatBoost	0.8492
3	Random Forest	0.8410
4	SVM	0.8250
5	MLP	0.8120

7. Explainability (SHAP)

SHAP (SHapley Additive exPlanations) values show the contribution of each feature to individual predictions. Positive values push predictions higher; negative values push them lower.

8. AI Insights

The following insights were generated by the CodeAlpha AI assistant using Retrieval-Augmented Generation (RAG) over your experiment history:

Performance Matrix Assessment

- The XGBoost ensemble outperforms other traditional and connectionist estimators with a composite accuracy of 85.25%.
- Tree-based ensembles (XGBoost, CatBoost, Random Forest) generalize better than SVM/MLP on this tabular clinical medical dataset.
- Features like Age, Cholesterol level, and Max Heart Rate have the highest clinical impact based on SHAP values.

9. Recommendations

1. Deploy XGBoost model as the primary production estimator for clinical decision support systems.

2. Monitor feature drift on cholesterol levels and blood pressure variables weekly.

3. Implement automated alerts for predictions with probability confidence scores above 75% to trigger immediate physician verification.

10. Future Work

- Expand the dataset with additional labelled samples to improve generalisation.
- Explore ensemble stacking of the top-3 leaderboard models.
- Apply neural architecture search (NAS) for deep learning tasks.
- Implement online learning for continuous model retraining on production data.
- Add fairness metrics (demographic parity, equalised odds) for regulated domains.
- Profile memory and compute efficiency for edge deployment.